

Efstratios Tsoukanis

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POSITIONS

- Visiting Scholar** Current
Institute of Mathematical Sciences, Claremont Graduate University, Claremont, CA
- Research with Distinguished Research Professor Hrushikesh Mhaskar on computational harmonic analysis and machine learning, supported by the US Office of Naval Research.
- Research Assistant Professor** 2024–2025
Institute of Mathematical Sciences, Claremont Graduate University, Claremont, CA

EDUCATION

- Ph.D. in Mathematics** 2024
University of Maryland, College Park
- Dissertation: *G-Invariant Representations using Coorbits*
 - Advisor: Professor Radu Balan
- M.S. in Mathematics** 2019
University of Crete, Heraklion
- Thesis: *Fourier multiplier and Fefferman counter-example*
 - Advisor: Professor Themistoklis Mitsis
- B.S. in Mathematics** 2016
University of Crete, Heraklion
- Erasmus exchange at the University of Barcelona, 2014

RESEARCH INTERESTS

Applied harmonic analysis and geometric machine learning: group-invariant representations (embeddings of orbit spaces, injectivity and bi-Lipschitz properties), phase retrieval, kernel methods and approximation theory.

PUBLICATIONS

Journal articles

1. R. Balan and **E. Tsoukanis**. G-invariant representations using coorbits: Bi-Lipschitz properties. *Advances in Computational Mathematics*, to appear, 2026. arXiv:2308.11784
2. N. Dym, M. Wellershoff, **E. Tsoukanis**, D. Levy, and R. Balan. Quantitative bounds for sorting-based permutation-invariant embeddings. *IEEE Transactions on Information Theory*, to appear, 2026. arXiv:2510.22186
3. H. N. Mhaskar, **E. Tsoukanis**, and A. D. Jagtap. An approximation theory perspective on machine learning. *Neural Networks*, 2026. ScienceDirect

Conference proceedings

4. H. N. Mhaskar, R. O'Dowd, and **E. Tsoukanis**. Active learning classification from a signal separation perspective. *International Conference on Sampling Theory and Applications (SampTA)*, Vienna, Austria, 2025. arXiv:2502.16425

5. R. Balan and E. Tsoukanis. Relationships between the phase retrieval problem and permutation invariant embeddings. *International Conference on Sampling Theory and Applications (SampTA)*, New Haven, CT, 2023. arXiv:2306.13111

Preprints

6. R. Balan, E. Tsoukanis, and M. Wellershoff. Stability of sorting based embeddings. arXiv:2410.05446, 2024.
7. R. Balan and E. Tsoukanis. G-invariant representations using coorbits: Injectivity properties. arXiv:2310.16365, 2023.

TALKS

- 2025 *Active learning classification from a signal separation perspective*. SampTA 2025, Vienna, Austria.
- 2023 *Relationships between the phase retrieval problem and permutation invariant embeddings*. SampTA 2023, Yale University, New Haven, CT.
- 2023 *Stability properties of coorbit embeddings*. CodEx Seminar.
- 2022 *Noise stability of functions with low influences*. Computational Learning Theory and Fourier Analysis Summer School.

TEACHING

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|---|-----------|
| Teaching Assistant, University of Maryland | 2018–2023 |
| Calculus II and III, Linear Algebra, Differential Equations | |
| Teaching Assistant, University of Crete | 2015–2016 |
| Calculus I, Analysis I | |

SERVICE AND OTHER ACTIVITIES

- Referee for *Foundations of Computational Mathematics* and *Applied and Computational Harmonic Analysis*
- Research Interaction Team in Applied Harmonic Analysis: Fall 2021, Fall 2023
- Research Interaction Team in Deep Learning: Fall 2022, Spring 2023, Fall 2023

SKILLS

Programming: Python, MATLAB, $\text{\LaTeX}$ **Languages:** Greek, English